

Introduction to Electronic Design Automation

Spring 2026

National Taiwan University

Problem Set 4

Due by 2026/06/10 23:59

1 [Partition]

[10%] Given an undirected graph $G = (V, E)$, let $V = \{a, b, c, d, e, f\}$ and E be the set of weighted edges as shown in the following table. Suppose that the vertices are partitioned into $\{a, b, c\}$ and $\{d, e, f\}$ initially. Apply the Kernighan-Lin algorithm to find an optimal balanced bi-partition and show intermediate steps.

	a	b	c	d	e	f
a	0	2	1	3	3	2
b	2	0	0	2	4	1
c	1	0	0	0	2	2
d	3	2	0	0	1	0
e	3	4	2	1	0	1
f	2	1	2	0	1	0

2 [Floorplanning]

[20%] Let expression $E = 123VH4H56VV78HH$.

module	width	height
1	2	5
2	3	1
3	4	3
4	2	2
5	4	7
6	5	4
7	3	4
8	6	3

- (a) (3%) Is E a Polish expression? Why or why not?
- (b) (8%) Is E a normalized Polish expression? List ALL normalized Polish expressions that express the same floorplan as E and sketch their slicing trees. Is the normalized Polish expression canonical in representing a floorplan?

- (c) (6%) Suppose that the modules are rigid and rotations are allowed. Determine the smallest bounding box that contains all the modules with respect to E . List all optimal solutions of each vertex in the slicing tree of E . Draw the final optimum floorplan.
- (d) (3%) In determining the smallest bounding box, can different Polish expressions of the same floorplan yield different answers? Why or why not?

3 [Wire-length Estimation]

[12%] Consider a net N with its four pins p_1 , p_2 , p_3 , and p_4 locating at coordinates $(1, 3)$, $(2, 5)$, $(4, 2)$, and $(6, 0)$, respectively. Suppose the pins are placed on a grid layout surface. Calculate the wire length of N by the following means:

- (a) (3%) semi-perimeter length,
- (b) (3%) complete graph length,
- (c) (3%) minimum spanning tree length, and
- (d) (3%) minimum Steiner tree length.

4 [Force-Directed Placement]

[8%] Consider a module A to be placed on a 10×10 grid layout, in which modules B , C , D , E , and F have been placed on grids $(0, 0)$, $(2, 5)$, $(6, 9)$, $(8, 1)$, and $(5, 4)$, respectively. Suppose A is connected to B , C , D , E , and F with weights 1, 3, 1, 2, and 3, respectively.

- (a) (4%) Find the location (grid) to place A by the force-directed placement. What is the corresponding weighted wire length?
- (b) (4%) Find the actual best location to place A , if it differs from that in (a), that minimizes the total weighted wire length. What is the corresponding weighted length?

(Rectilinear connections are assumed in the final layout.)

5 [Maze Routing]

[18%] Given the layout of Fig. 1, where the gray grids denote obstacles.

- (a) (3%) Find a shortest path between S and U by Lee's approach. Specify the propagation and retrace steps. What is the path length?
- (b) (6%) Find a shortest path between S and T by Akers's approach (including Way 1 and Way 2). Specify the propagation and retrace steps. What is the path length?
- (c) (3%) Find a shortest path between U to T by Hadlock's approach. Specify the propagation and retrace steps. What is the path length?
- (d) (3%) Connect pins S , T , and U by finding the minimum rectilinear spanning tree. What is the total net length?
- (e) (3%) Connect pins S , T , and U by finding the minimum rectilinear Steiner tree. Identify the Steiner point. What is the total net length?

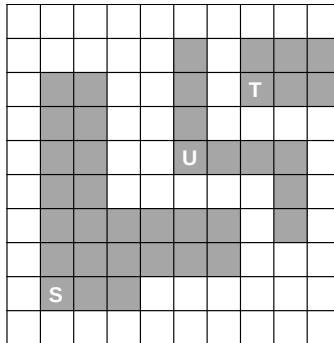


Fig. 1. Layout for maze routing.

6 [Channel Routing]

[18%] Consider a channel with ordered pins $(1, 2, 2, 4, 3, 6, \emptyset, 5, 6)$ on the upper boundary and $(2, \emptyset, 3, 7, 1, 3, 7, 4, 5)$ on the lower boundary, respectively, where “ $\emptyset$ ” denotes empty pin. Assume the reserved HV layer model (i.e., metal layer 1 and metal layer 2 reversed for horizontal and vertical routing, respectively).

- (6%) Draw the horizontal and vertical constraint graphs of the channel routing problem.
- (6%) Is the basic left-edge algorithm applicable to this routing instance? Is the constrained left-edge algorithm applicable to this instance? Why or why not? If yes, give the routing result.
- (6%) Perform routing by the constrained left-edge algorithm with doglegging. Show your derivation steps.

7 [Testing]

[14%] By fault equivalence and dominance, identify the prime faults of the circuit of Figure 2. What are sets of tests of these prime faults?

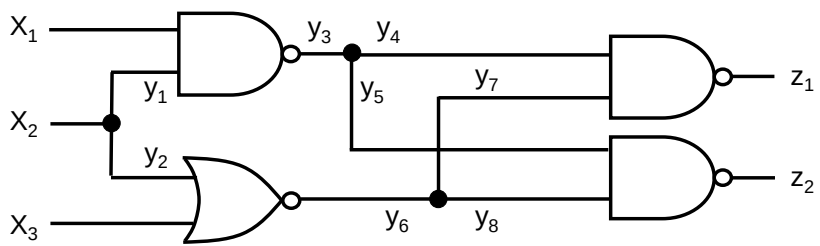


Fig. 2. A circuit under stuck-at fault analysis.